Tangerine_part4_Bayesian_analysis

Tangerine

Buddy challenge part 4: Bayesian models

Seting up working directory, packages and data

```
# just run this code chunk
# function to ignoring the setting of the relative path below when knitting
run_if_not_knitting <- function(expr) {
  if (!isTRUE(getOption("knitr.in.progress"))) {
    eval(expr)
  }
}

# nifty code using the pacman package
# it checks if the packages specified below are installed, if not,
#they will be installed, if yes, they will be loaded
if (!require("pacman")) install.packages("pacman")
```

```
Loading required package: pacman
```

```
pacman::p_load(rstudioapi, tidyverse, bayesrules,
               tidybayes, trimr, brms, bayesplot, bayestestR, modelr, lme4)

# set the current working directory to the one where this file is
run_if_not_knitting(current_working_dir <-
                      dirname(rstudioapi::getActiveDocumentContext()$path))
run_if_not_knitting(setwd(current_working_dir))

# options to increase efficiency of brms models (optional)
#options(brms.backend = "cmdstanr")
options(mc.cores = parallel::detectCores())
```

```
Rows: 2880 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): pic, sound, condition
dbl (5): participant, run, rt, accuracy, SOA

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
[1] "Categorical" "Mediated"
```

```
[1] "related"   "unrelated"
```

Bayesian model closest to Mascelloni et al. (2021) with flat (ish) priors

```r
# Copy pasting the Mascelloni et al maximal model that did NOT converge
# lmer.exp1 <- lmer(rt ~ distractor_type * distractor_relatedness +
                   # (1 + distractor_type.num *
                    #   distractor_relatedness.num | participant) +
                  #  (1 + distractor_type.num *
                   #    distractor_relatedness.num | pic) +
                  #  (1 + distractor_relatedness.num | distractor_word),
                  # data=exp1_trimmedData,
                  # control=lmerControl(optimizer="bobyqa"),
                  # verbose=2)

# Getting our priors
maximal <- bf(rt ~ distractor_type * distractor_relatedness +
                    (1 + distractor_type *
                       distractor_relatedness | participant) +
                    (1 + distractor_type *
                       distractor_relatedness | pic) +
                    (1 + distractor_relatedness | distractor_word))

get_prior(maximal, aud, family = "gaussian")
```

```
                     prior     class                                     coef
                    (flat)         b
                    (flat)         b                  distractor_relatedness1
                    (flat)         b                         distractor_type1
```

```
                    (flat)         b distractor_type1:distractor_relatedness1
                    lkj(1)       cor
                    lkj(1)       cor
                    lkj(1)       cor
                    lkj(1)       cor
student_t(3, 718.4, 137.6) Intercept
    student_t(3, 0, 137.6)        sd
    student_t(3, 0, 137.6)        sd
    student_t(3, 0, 137.6)        sd                  distractor_relatedness1
    student_t(3, 0, 137.6)        sd                                Intercept
    student_t(3, 0, 137.6)        sd
    student_t(3, 0, 137.6)        sd                  distractor_relatedness1
    student_t(3, 0, 137.6)        sd                         distractor_type1
    student_t(3, 0, 137.6)        sd distractor_type1:distractor_relatedness1
    student_t(3, 0, 137.6)        sd                                Intercept
    student_t(3, 0, 137.6)        sd
    student_t(3, 0, 137.6)        sd                  distractor_relatedness1
    student_t(3, 0, 137.6)        sd                         distractor_type1
    student_t(3, 0, 137.6)        sd distractor_type1:distractor_relatedness1
    student_t(3, 0, 137.6)        sd                                Intercept
    student_t(3, 0, 137.6)     sigma
          group resp dpar nlpar lb ub tag       source
                                              default
                                        (vectorized)
                                        (vectorized)
                                        (vectorized)
                                              default
distractor_word                         (vectorized)
    participant                         (vectorized)
            pic                         (vectorized)
                                              default
                                 0            default
distractor_word                  0      (vectorized)
distractor_word                  0      (vectorized)
distractor_word                  0      (vectorized)
    participant                  0      (vectorized)
    participant                  0      (vectorized)
    participant                  0      (vectorized)
    participant                  0      (vectorized)
    participant                  0      (vectorized)
            pic                  0      (vectorized)
            pic                  0      (vectorized)
            pic                  0      (vectorized)
```

```
            pic                  0        (vectorized)
            pic                  0        (vectorized)
                                 0             default
```

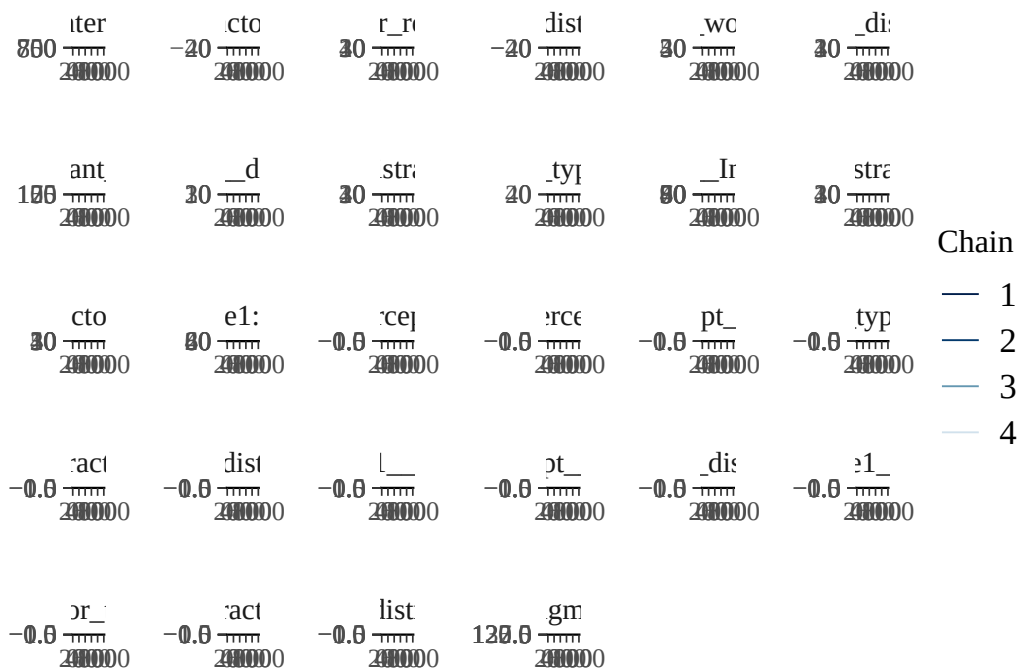
```r
# Setting flat priors

# We're aware we're not technically setting completely flat priors since,
# as McElreath points out, flat means all values are possible, but we need
# the computer to work and also negative rt values are absurd

flat_priors <- c(prior(uniform(0, 1800), class = "Intercept"),
                 #not putting anything for class b since the model will
                 #automatically do flat if not told otherwise
                 #we put a broad prior for the sd and sigma which is the closest
                 #we manage to get to fully flat with our machines being able
                 #to run this. We initially had way flatter but it kept "frying"
                 #the computer we were using
               prior(normal(0, 50), class = "sd"),
               prior(normal(0, 50), class = "sigma"))


flat_model_bayesian <- brm(
  formula = maximal,
  data = aud,
  prior = flat_priors,
  family = gaussian(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  seed = 42,
  thin = 1,
  cores = 4
)
```

```
Warning: It appears as if you have specified a lower bounded prior on a parameter that has no
If this is really what you want, please specify argument 'lb' of 'set_prior' appropriately.
Warning occurred for prior
Intercept ~ uniform(0, 1800)

Warning: It appears as if you have specified an upper bounded prior on a parameter that has n
If this is really what you want, please specify argument 'ub' of 'set_prior' appropriately.
Warning occurred for prior
```

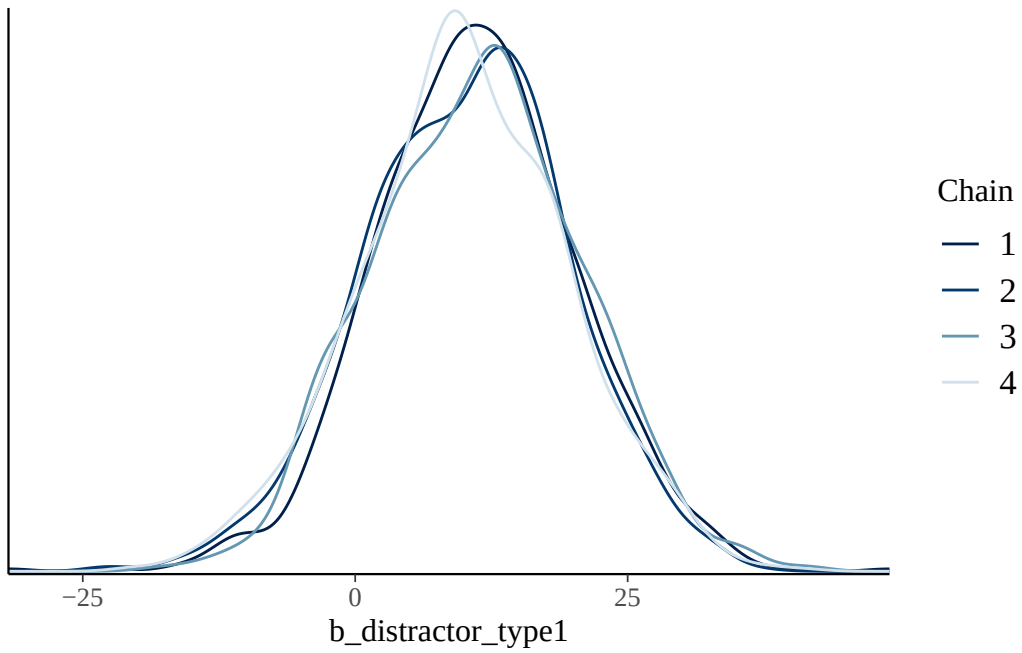
```
Intercept ~ uniform(0, 1800)

Compiling Stan program...

Start sampling
```

```r
# checking convergence diagnostics
mcmc_plot(flat_model_bayesian,
          type = 'trace')
```

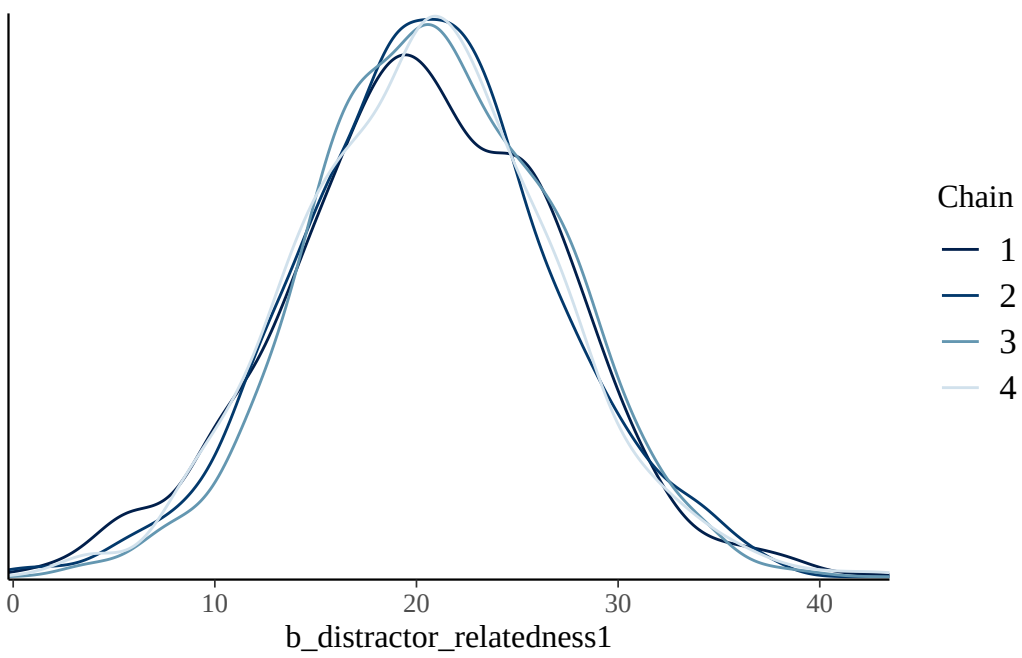
```
No divergences to plot.
```



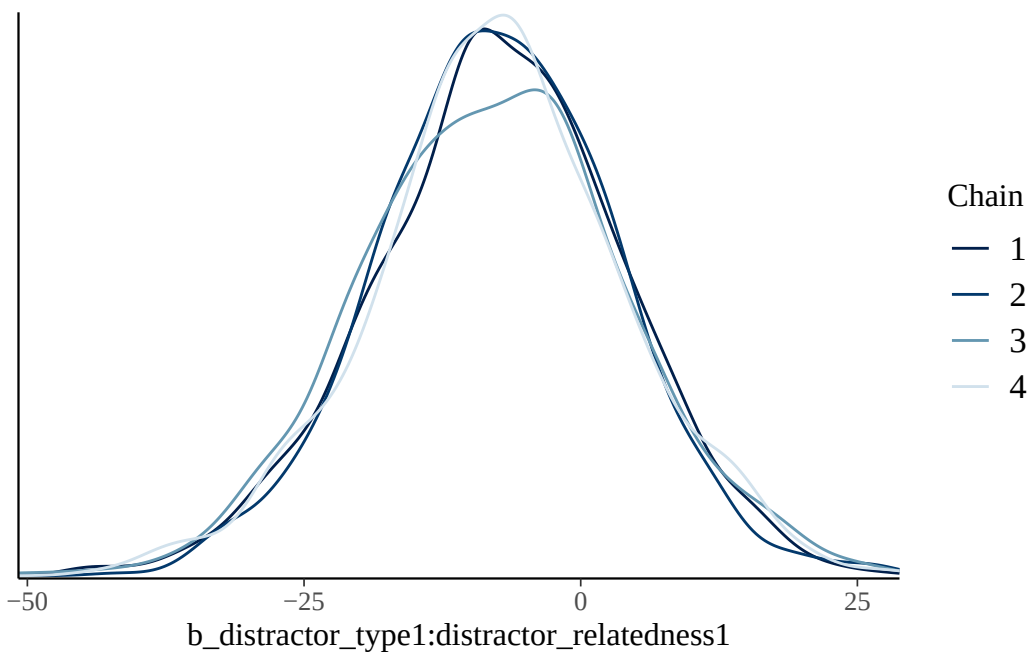

```r
# looks like healthy caterpillars
mcmc_dens_overlay(flat_model_bayesian,
                  pars = "b_distractor_type1")
```




```
mcmc_dens_overlay(flat_model_bayesian,
                  pars = "b_distractor_relatedness1")
```



```r
mcmc_dens_overlay(flat_model_bayesian,
                  pars = "b_distractor_type1:distractor_relatedness1")
```




```r
# Overlay for the fixed effects estimates look good too

# looking at model output
flat_model_bayesian
```

```
 Family: gaussian
  Links: mu = identity
Formula: rt ~ distractor_type * distractor_relatedness + (1 + distractor_type * distractor_re
   Data: aud (Number of observations: 2765)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~distractor_word (Number of levels: 72)
                                      Estimate Est.Error l-95% CI u-95% CI
sd(Intercept)                            32.60      4.83    23.14    42.44
sd(distractor_relatedness1)               9.95      7.31     0.40    27.29
cor(Intercept,distractor_relatedness1)   -0.30      0.52    -0.97     0.82
                                      Rhat Bulk_ESS Tail_ESS
```

```
sd(Intercept)                           1.00      997     1403
sd(distractor_relatedness1)             1.00     1181     1728
cor(Intercept,distractor_relatedness1) 1.00     2408     2395

~participant (Number of levels: 20)
                                                                      Estimate
sd(Intercept)                                                            92.75
sd(distractor_type1)                                                      6.47
sd(distractor_relatedness1)                                              10.76
sd(distractor_type1:distractor_relatedness1)                             11.36
cor(Intercept,distractor_type1)                                          -0.11
cor(Intercept,distractor_relatedness1)                                    0.18
cor(distractor_type1,distractor_relatedness1)                            -0.12
cor(Intercept,distractor_type1:distractor_relatedness1)                   0.14
cor(distractor_type1,distractor_type1:distractor_relatedness1)           -0.02
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    -0.05
                                                                      Est.Error
sd(Intercept)                                                             13.98
sd(distractor_type1)                                                       5.03
sd(distractor_relatedness1)                                                6.90
sd(distractor_type1:distractor_relatedness1)                               8.47
cor(Intercept,distractor_type1)                                            0.43
cor(Intercept,distractor_relatedness1)                                     0.36
cor(distractor_type1,distractor_relatedness1)                              0.44
cor(Intercept,distractor_type1:distractor_relatedness1)                    0.44
cor(distractor_type1,distractor_type1:distractor_relatedness1)             0.44
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)      0.44
                                                                      l-95% CI
sd(Intercept)                                                            69.22
sd(distractor_type1)                                                      0.27
sd(distractor_relatedness1)                                               0.50
sd(distractor_type1:distractor_relatedness1)                              0.63
cor(Intercept,distractor_type1)                                          -0.84
cor(Intercept,distractor_relatedness1)                                   -0.57
cor(distractor_type1,distractor_relatedness1)                            -0.86
cor(Intercept,distractor_type1:distractor_relatedness1)                  -0.76
cor(distractor_type1,distractor_type1:distractor_relatedness1)           -0.82
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    -0.83
                                                                      u-95% CI
sd(Intercept)                                                           123.95
sd(distractor_type1)                                                     18.69
sd(distractor_relatedness1)                                              25.59
sd(distractor_type1:distractor_relatedness1)                             31.45
```

```
cor(Intercept,distractor_type1)                                          0.73
cor(Intercept,distractor_relatedness1)                                   0.81
cor(distractor_type1,distractor_relatedness1)                            0.75
cor(Intercept,distractor_type1:distractor_relatedness1)                  0.85
cor(distractor_type1,distractor_type1:distractor_relatedness1)           0.79
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    0.77
                                                                       Rhat
sd(Intercept)                                                          1.00
sd(distractor_type1)                                                   1.00
sd(distractor_relatedness1)                                            1.00
sd(distractor_type1:distractor_relatedness1)                           1.00
cor(Intercept,distractor_type1)                                        1.00
cor(Intercept,distractor_relatedness1)                                 1.00
cor(distractor_type1,distractor_relatedness1)                          1.00
cor(Intercept,distractor_type1:distractor_relatedness1)                1.00
cor(distractor_type1,distractor_type1:distractor_relatedness1)         1.00
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1) 1.00
                                                                       Bulk_ESS
sd(Intercept)                                                           843
sd(distractor_type1)                                                   2003
sd(distractor_relatedness1)                                            1401
sd(distractor_type1:distractor_relatedness1)                           2698
cor(Intercept,distractor_type1)                                        4453
cor(Intercept,distractor_relatedness1)                                 4708
cor(distractor_type1,distractor_relatedness1)                          2347
cor(Intercept,distractor_type1:distractor_relatedness1)                5247
cor(distractor_type1,distractor_type1:distractor_relatedness1)         3733
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)  3394
                                                                       Tail_ESS
sd(Intercept)                                                          1265
sd(distractor_type1)                                                   2000
sd(distractor_relatedness1)                                            1652
sd(distractor_type1:distractor_relatedness1)                           2330
cor(Intercept,distractor_type1)                                        2544
cor(Intercept,distractor_relatedness1)                                 2913
cor(distractor_type1,distractor_relatedness1)                          2799
cor(Intercept,distractor_type1:distractor_relatedness1)                2592
cor(distractor_type1,distractor_type1:distractor_relatedness1)         2945
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)  3086

~pic (Number of levels: 36)
                                                                       Estimate
sd(Intercept)                                                             51.02
```

```
sd(distractor_type1)                                                    8.16
sd(distractor_relatedness1)                                            13.21
sd(distractor_type1:distractor_relatedness1)                           18.40
cor(Intercept,distractor_type1)                                        -0.08
cor(Intercept,distractor_relatedness1)                                 -0.12
cor(distractor_type1,distractor_relatedness1)                           0.02
cor(Intercept,distractor_type1:distractor_relatedness1)                 0.04
cor(distractor_type1,distractor_type1:distractor_relatedness1)         -0.02
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)   0.12
                                                                    Est.Error
sd(Intercept)                                                            7.34
sd(distractor_type1)                                                     6.19
sd(distractor_relatedness1)                                              9.33
sd(distractor_type1:distractor_relatedness1)                            14.09
cor(Intercept,distractor_type1)                                          0.42
cor(Intercept,distractor_relatedness1)                                   0.37
cor(distractor_type1,distractor_relatedness1)                            0.45
cor(Intercept,distractor_type1:distractor_relatedness1)                  0.41
cor(distractor_type1,distractor_type1:distractor_relatedness1)           0.45
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    0.46
                                                                    l-95% CI
sd(Intercept)                                                          38.73
sd(distractor_type1)                                                    0.30
sd(distractor_relatedness1)                                             0.54
sd(distractor_type1:distractor_relatedness1)                            0.63
cor(Intercept,distractor_type1)                                        -0.81
cor(Intercept,distractor_relatedness1)                                 -0.79
cor(distractor_type1,distractor_relatedness1)                          -0.81
cor(Intercept,distractor_type1:distractor_relatedness1)                -0.76
cor(distractor_type1,distractor_type1:distractor_relatedness1)         -0.82
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)  -0.77
                                                                    u-95% CI
sd(Intercept)                                                          67.07
sd(distractor_type1)                                                   23.29
sd(distractor_relatedness1)                                            34.54
sd(distractor_type1:distractor_relatedness1)                           52.20
cor(Intercept,distractor_type1)                                         0.73
cor(Intercept,distractor_relatedness1)                                  0.67
cor(distractor_type1,distractor_relatedness1)                           0.81
cor(Intercept,distractor_type1:distractor_relatedness1)                 0.79
cor(distractor_type1,distractor_type1:distractor_relatedness1)          0.81
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)   0.88
                                                                    Rhat
```

```
sd(Intercept)                                                          1.00
sd(distractor_type1)                                                   1.00
sd(distractor_relatedness1)                                            1.00
sd(distractor_type1:distractor_relatedness1)                           1.00
cor(Intercept,distractor_type1)                                        1.00
cor(Intercept,distractor_relatedness1)                                 1.00
cor(distractor_type1,distractor_relatedness1)                          1.00
cor(Intercept,distractor_type1:distractor_relatedness1)                1.00
cor(distractor_type1,distractor_type1:distractor_relatedness1)         1.00
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1) 1.00
                                                                       Bulk_ESS
sd(Intercept)                                                              1227
sd(distractor_type1)                                                       1884
sd(distractor_relatedness1)                                                 741
sd(distractor_type1:distractor_relatedness1)                               1090
cor(Intercept,distractor_type1)                                            4461
cor(Intercept,distractor_relatedness1)                                     3184
cor(distractor_type1,distractor_relatedness1)                              1644
cor(Intercept,distractor_type1:distractor_relatedness1)                    4265
cor(distractor_type1,distractor_type1:distractor_relatedness1)             2591
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)      1863
                                                                       Tail_ESS
sd(Intercept)                                                              2048
sd(distractor_type1)                                                       2355
sd(distractor_relatedness1)                                                1829
sd(distractor_type1:distractor_relatedness1)                               1633
cor(Intercept,distractor_type1)                                            3041
cor(Intercept,distractor_relatedness1)                                     2744
cor(distractor_type1,distractor_relatedness1)                              2373
cor(Intercept,distractor_type1:distractor_relatedness1)                    2814
cor(distractor_type1,distractor_type1:distractor_relatedness1)             3092
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)      2684

Regression Coefficients:
                                        Estimate Est.Error l-95% CI u-95% CI
Intercept                                 754.40     23.48   707.63   800.37
distractor_type1                           10.67      9.36    -7.72    28.88
distractor_relatedness1                    20.57      6.29     8.03    33.06
distractor_type1:distractor_relatedness1   -7.57     11.25   -30.17    14.84
                                        Rhat Bulk_ESS Tail_ESS
Intercept                               1.00      505      625
distractor_type1                        1.00     1545     2670
distractor_relatedness1                 1.00     3582     3034
```

```
distractor_type1:distractor_relatedness1 1.00     3920     2735

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   123.97      1.69   120.73   127.23 1.00     5353     2970

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
# Rhat and ESS values are fine

# probability of direction
p_direction(flat_model_bayesian)
```

```
Probability of Direction

Parameter                                |     pd
--------------------------------------------------
(Intercept)                              |   100%
distractor_type1                         | 87.40%
distractor_relatedness1                  | 99.98%
distractor_type1:distractor_relatedness1 | 75.58%
```

```r
# computing p-value out of pd of direction so we can compare it to the original
# paper (we are aware that we are not actually using flat priors and the values
# we get therefore cant actually be trusted for comparison)

# p-value for b_distractor_type1
pd_to_p(pd = 0.8688, direction = 'two-sided')
```

```
[1] 0.2624
```

```r
# p-value: 0.2624

# p-value for b_distractor_relatedness1
pd_to_p(pd = 0.9992, direction = 'two-sided')
```

```
[1] 0.0016
```

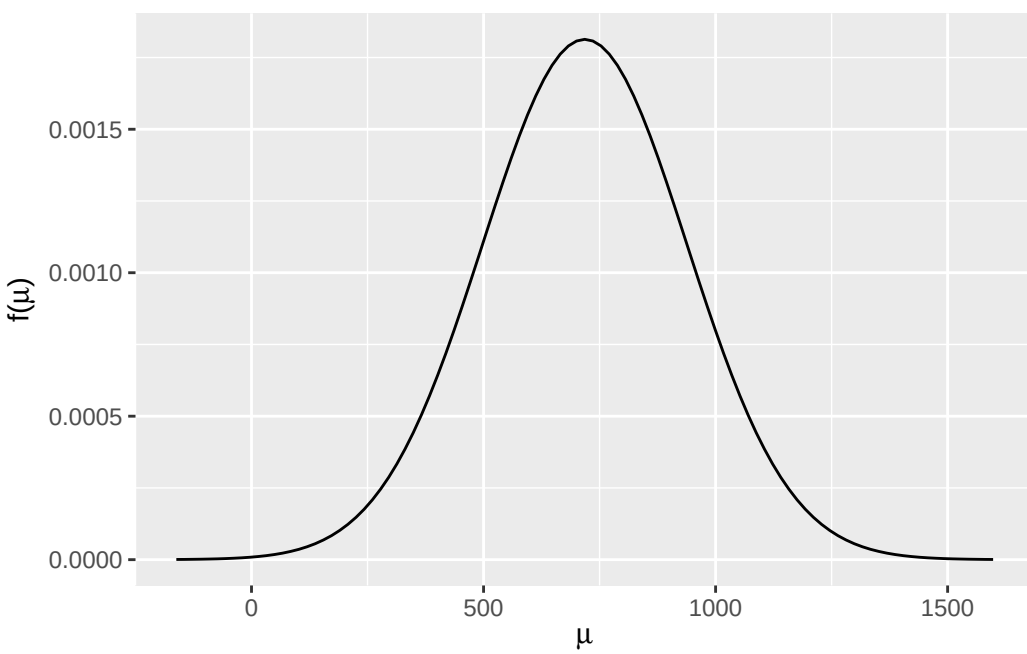
```r
# p-value: 0.0016

# p-value for b_distractor_type1:distractor_relatedness1
pd_to_p(pd = 0.7502, direction = 'two-sided')
```

```
[1] 0.4996
```

```r
# p-value: 0.4996


# In the original analysis with the reduced model, they report p-values of 0.252
# for distractor type, 0.001 for relatedness and 0.428 for the interaction term.
# These values are quite similar to what we got and we get the exact same
# "significance" which is only relatedness.
```

Bayesian model with weakly informative priors

```r
# We tested things with this plot and landed on the following numbers:
# Reasoning was to keep things wide and "open"
plot_normal(718, 220)
```

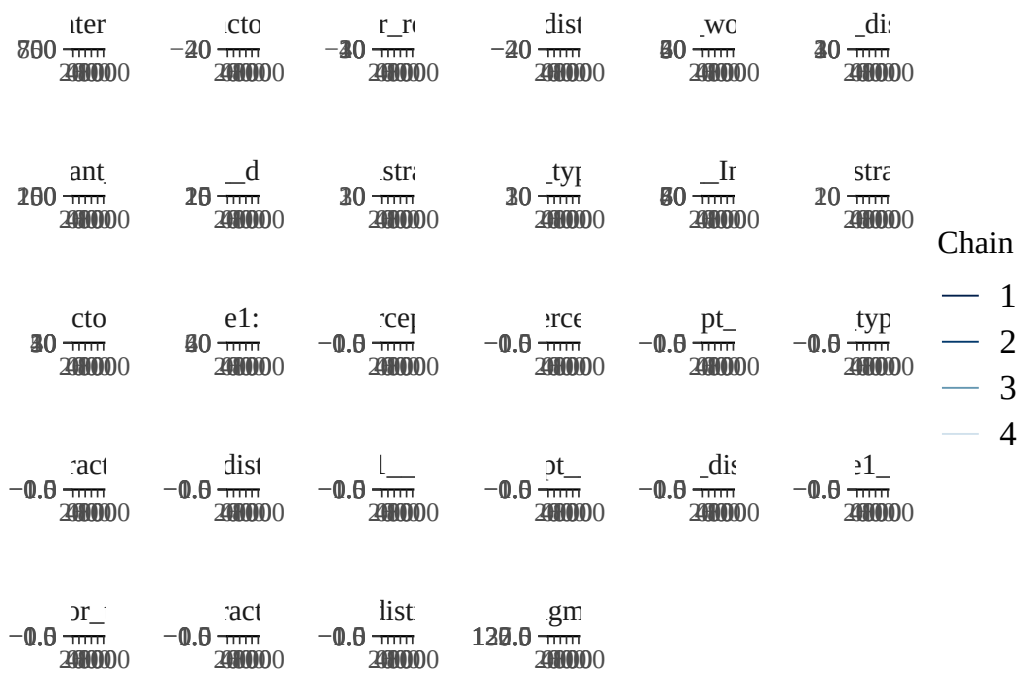


```r
# Setting weakly informative priors, taken from the normal ploting we just did
#and from R's "get_priors"'s "advice"
weak_priors <- c(prior(normal(718, 220), class = "Intercept"),
                 prior(normal(0, 220), class = "b",
                       coef = "distractor_relatedness1"),
              prior(normal(0, 220), class = "b",
                    coef = "distractor_type1"),
              prior(normal(0, 220), class = "b",
                    coef = "distractor_type1:distractor_relatedness1"),
              prior(lkj(1), class = 'cor'),
              prior(cauchy(0, 3), class = 'sd', group = 'distractor_word'),
              prior(cauchy(0, 3), class = 'sd', group = 'participant'),
              prior(cauchy(0, 3), class = 'sd', group = 'pic'),
              prior(normal(0, 138), class = "sigma"))


proper_model <- brm(
  formula = maximal,
  data = aud,
  prior = weak_priors,
  family = gaussian(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  thin = 1,
  seed = 42,
  cores = 4
)
```

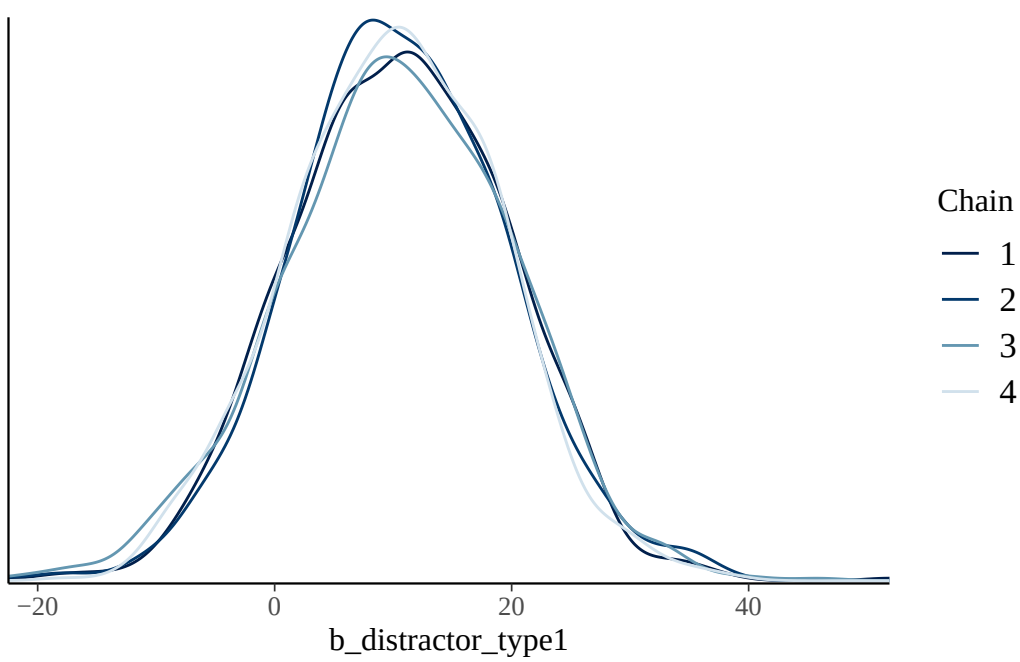
```
Compiling Stan program...

Start sampling
```

```r
# checking convergence diagnostics
mcmc_plot(proper_model,
          type = 'trace')
```

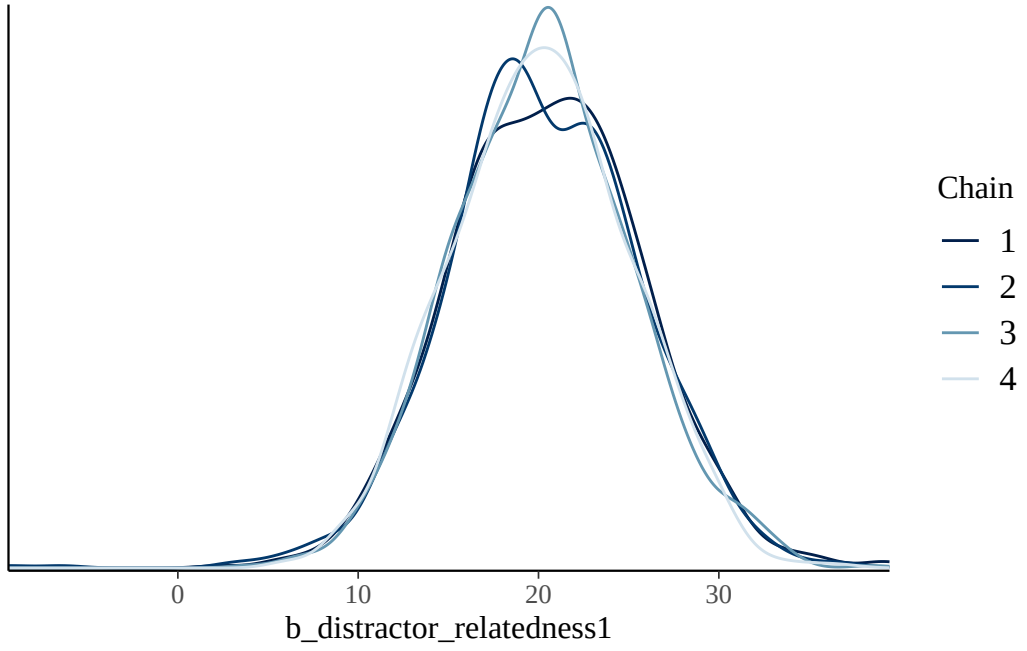
```
No divergences to plot.
```



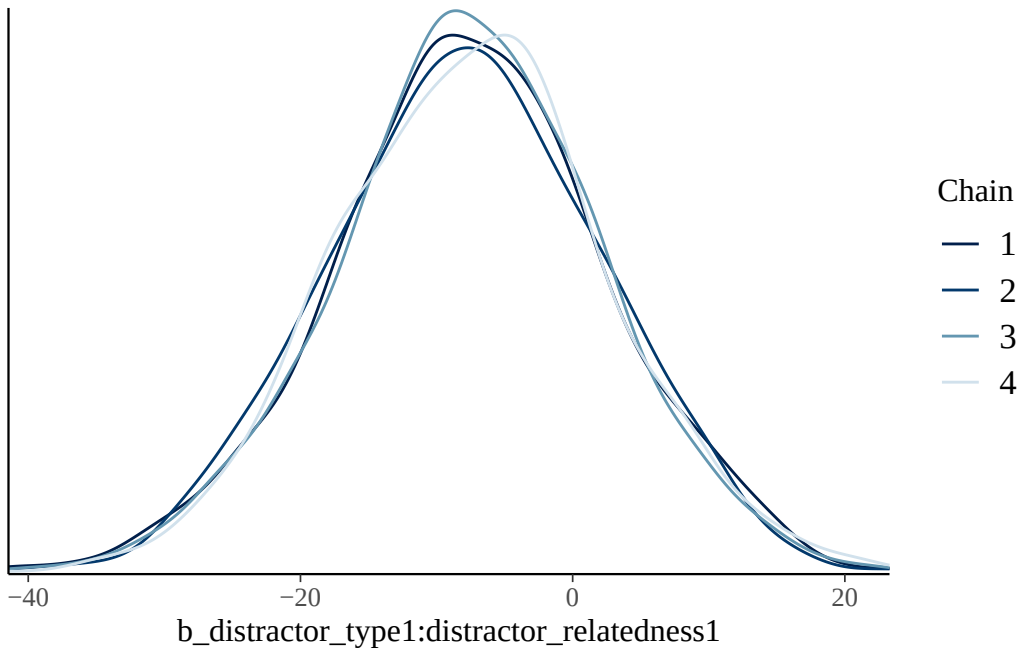

```
# looks like healthy caterpillars
mcmc_dens_overlay(proper_model,
                  pars = "b_distractor_type1")
```



```r
mcmc_dens_overlay(proper_model,
                  pars = "b_distractor_relatedness1")
```




```r
mcmc_dens_overlay(proper_model,
                  pars = "b_distractor_type1:distractor_relatedness1")
```




```
# Overlay for the fixed effects estimates look good too

# looking at model output
proper_model
```

```
    16.16
sd(distractor_type1:distractor_relatedness1)                              15.22
cor(Intercept,distractor_type1)                                            0.75
cor(Intercept,distractor_relatedness1)                                     0.82
cor(distractor_type1,distractor_relatedness1)                              0.79
cor(Intercept,distractor_type1:distractor_relatedness1)                    0.82
cor(distractor_type1,distractor_type1:distractor_relatedness1)             0.80
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)      0.82
                                                                        Rhat
sd(Intercept)                                                           1.00
sd(distractor_type1)                                                    1.00
sd(distractor_relatedness1)                                             1.00
sd(distractor_type1:distractor_relatedness1)                            1.00
cor(Intercept,distractor_type1)                                         1.00
cor(Intercept,distractor_relatedness1)                                  1.00
cor(distractor_type1,distractor_relatedness1)                           1.00
cor(Intercept,distractor_type1:distractor_relatedness1)                 1.00
```

```
cor(distractor_type1,distractor_type1:distractor_relatedness1)        1.00
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1) 1.00
                                                                      Bulk_ESS
sd(Intercept)                                                             1196
sd(distractor_type1)                                                      3274
sd(distractor_relatedness1)                                               2270
sd(distractor_type1:distractor_relatedness1)                              5148
cor(Intercept,distractor_type1)                                           7003
cor(Intercept,distractor_relatedness1)                                    6400
cor(distractor_type1,distractor_relatedness1)                             3813
cor(Intercept,distractor_type1:distractor_relatedness1)                   8256
cor(distractor_type1,distractor_type1:distractor_relatedness1)            4378
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     3664
                                                                      Tail_ESS
sd(Intercept)                                                             1877
sd(distractor_type1)                                                      2285
sd(distractor_relatedness1)                                               2232
sd(distractor_type1:distractor_relatedness1)                              2538
cor(Intercept,distractor_type1)                                           2739
cor(Intercept,distractor_relatedness1)                                    2283
cor(distractor_type1,distractor_relatedness1)                             2870
cor(Intercept,distractor_type1:distractor_relatedness1)                   2522
cor(distractor_type1,distractor_type1:distractor_relatedness1)            2769
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     3027

~pic (Number of levels: 36)
                                                                      Estimate
sd(Intercept)                                                            50.48
sd(distractor_type1)                                                      3.29
sd(distractor_relatedness1)                                               4.05
sd(distractor_type1:distractor_relatedness1)                              4.66
cor(Intercept,distractor_type1)                                          -0.02
cor(Intercept,distractor_relatedness1)                                   -0.07
cor(distractor_type1,distractor_relatedness1)                             0.02
cor(Intercept,distractor_type1:distractor_relatedness1)                   0.01
cor(distractor_type1,distractor_type1:distractor_relatedness1)            0.01
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     0.01
                                                                      Est.Error
sd(Intercept)                                                              7.04
sd(distractor_type1)                                                       3.33
sd(distractor_relatedness1)                                                4.85
sd(distractor_type1:distractor_relatedness1)                               6.02
cor(Intercept,distractor_type1)                                            0.44
```

```
cor(Intercept,distractor_relatedness1)                                    0.42
cor(distractor_type1,distractor_relatedness1)                             0.45
cor(Intercept,distractor_type1:distractor_relatedness1)                   0.44
cor(distractor_type1,distractor_type1:distractor_relatedness1)            0.45
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     0.45
                                                                     l-95% CI
sd(Intercept)                                                           38.55
sd(distractor_type1)                                                     0.10
sd(distractor_relatedness1)                                              0.13
sd(distractor_type1:distractor_relatedness1)                             0.09
cor(Intercept,distractor_type1)                                         -0.80
cor(Intercept,distractor_relatedness1)                                  -0.80
cor(distractor_type1,distractor_relatedness1)                           -0.80
cor(Intercept,distractor_type1:distractor_relatedness1)                 -0.79
cor(distractor_type1,distractor_type1:distractor_relatedness1)          -0.81
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)   -0.82
                                                                     u-95% CI
sd(Intercept)                                                           66.26
sd(distractor_type1)                                                    12.45
sd(distractor_relatedness1)                                             17.31
sd(distractor_type1:distractor_relatedness1)                            22.23
cor(Intercept,distractor_type1)                                          0.79
cor(Intercept,distractor_relatedness1)                                   0.76
cor(distractor_type1,distractor_relatedness1)                            0.83
cor(Intercept,distractor_type1:distractor_relatedness1)                  0.81
cor(distractor_type1,distractor_type1:distractor_relatedness1)           0.82
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    0.82
                                                                     Rhat
sd(Intercept)                                                        1.00
sd(distractor_type1)                                                 1.00
sd(distractor_relatedness1)                                          1.00
sd(distractor_type1:distractor_relatedness1)                         1.00
cor(Intercept,distractor_type1)                                      1.00
cor(Intercept,distractor_relatedness1)                               1.00
cor(distractor_type1,distractor_relatedness1)                        1.00
cor(Intercept,distractor_type1:distractor_relatedness1)              1.00
cor(distractor_type1,distractor_type1:distractor_relatedness1)       1.00
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1) 1.00
                                                                     Bulk_ESS
sd(Intercept)                                                            1416
sd(distractor_type1)                                                     3063
sd(distractor_relatedness1)                                              2460
sd(distractor_type1:distractor_relatedness1)                             3354
```

```
cor(Intercept,distractor_type1)                                          5605
cor(Intercept,distractor_relatedness1)                                   6052
cor(distractor_type1,distractor_relatedness1)                            4216
cor(Intercept,distractor_type1:distractor_relatedness1)                  7651
cor(distractor_type1,distractor_type1:distractor_relatedness1)           4811
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    2911
                                                                      Tail_ESS
sd(Intercept)                                                            2411
sd(distractor_type1)                                                     1975
sd(distractor_relatedness1)                                              2119
sd(distractor_type1:distractor_relatedness1)                             2042
cor(Intercept,distractor_type1)                                          2823
cor(Intercept,distractor_relatedness1)                                   2960
cor(distractor_type1,distractor_relatedness1)                            3400
cor(Intercept,distractor_type1:distractor_relatedness1)                  2806
cor(distractor_type1,distractor_type1:distractor_relatedness1)           3033
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    3190

Regression Coefficients:
                                        Estimate Est.Error l-95% CI u-95% CI
Intercept                                 754.98     23.31   709.81   799.08
distractor_type1                           10.47      9.13    -7.42    27.92
distractor_relatedness1                    20.30      5.10    10.67    30.34
distractor_type1:distractor_relatedness1   -7.63      9.81   -27.12    11.64
                                        Rhat Bulk_ESS Tail_ESS
Intercept                               1.01      718     1450
distractor_type1                        1.00     3230     3089
distractor_relatedness1                 1.00     5471     3110
distractor_type1:distractor_relatedness1 1.00     6702     2973

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   124.12      1.69   120.78   127.59 1.00     6086     2768

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
# Rhat and ESS values are fine
```

Interpreting the results

```r
summary(proper_model)
```

```
 Family: gaussian
  Links: mu = identity
Formula: rt ~ distractor_type * distractor_relatedness + (1 + distractor_type * distractor_re
   Data: aud (Number of observations: 2765)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~distractor_word (Number of levels: 72)
                                       Estimate Est.Error l-95% CI u-95% CI
sd(Intercept)                             33.24      4.36    25.26    42.44
sd(distractor_relatedness1)                3.64      4.00     0.08    14.91
cor(Intercept,distractor_relatedness1)    -0.13      0.56    -0.96     0.91
                                       Rhat Bulk_ESS Tail_ESS
sd(Intercept)                          1.00     1894     3037
sd(distractor_relatedness1)            1.00     2652     2290
cor(Intercept,distractor_relatedness1) 1.00     4454     2477

~participant (Number of levels: 20)
                                                                   Estimate
sd(Intercept)                                                         95.95
sd(distractor_type1)                                                   2.93
sd(distractor_relatedness1)                                            4.31
sd(distractor_type1:distractor_relatedness1)                           3.79
cor(Intercept,distractor_type1)                                       -0.07
cor(Intercept,distractor_relatedness1)                                 0.12
cor(distractor_type1,distractor_relatedness1)                         -0.04
cor(Intercept,distractor_type1:distractor_relatedness1)                0.05
cor(distractor_type1,distractor_type1:distractor_relatedness1)        -0.01
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    -0.01
                                                                   Est.Error
sd(Intercept)                                                          16.37
sd(distractor_type1)                                                    2.84
sd(distractor_relatedness1)                                             4.36
sd(distractor_type1:distractor_relatedness1)                            4.18
cor(Intercept,distractor_type1)                                         0.44
cor(Intercept,distractor_relatedness1)                                  0.42
```

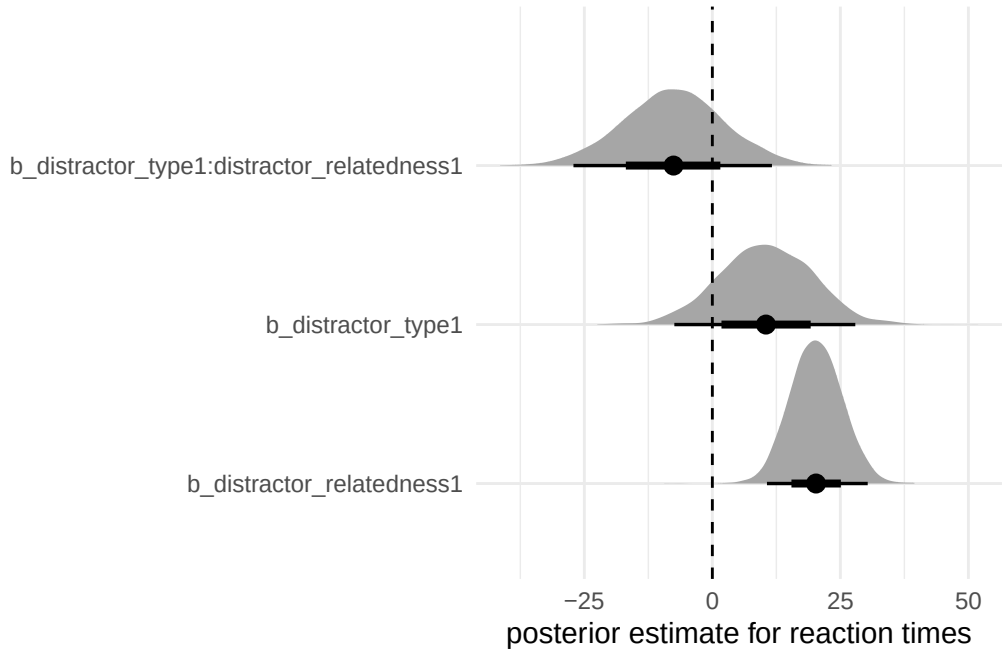
```
cor(distractor_type1,distractor_relatedness1)                              0.44
cor(Intercept,distractor_type1:distractor_relatedness1)                    0.45
cor(distractor_type1,distractor_type1:distractor_relatedness1)             0.44
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)      0.45
                                                                      l-95% CI
sd(Intercept)                                                            70.18
sd(distractor_type1)                                                      0.09
sd(distractor_relatedness1)                                               0.11
sd(distractor_type1:distractor_relatedness1)                              0.11
cor(Intercept,distractor_type1)                                          -0.83
cor(Intercept,distractor_relatedness1)                                   -0.72
cor(distractor_type1,distractor_relatedness1)                            -0.83
cor(Intercept,distractor_type1:distractor_relatedness1)                  -0.79
cor(distractor_type1,distractor_type1:distractor_relatedness1)           -0.81
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    -0.82
                                                                      u-95% CI
sd(Intercept)                                                           133.53
sd(distractor_type1)                                                     10.51
sd(distractor_relatedness1)                                              16.16
sd(distractor_type1:distractor_relatedness1)                             15.22
cor(Intercept,distractor_type1)                                           0.75
cor(Intercept,distractor_relatedness1)                                    0.82
cor(distractor_type1,distractor_relatedness1)                             0.79
cor(Intercept,distractor_type1:distractor_relatedness1)                   0.82
cor(distractor_type1,distractor_type1:distractor_relatedness1)            0.80
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     0.82
                                                                      Rhat
sd(Intercept)                                                         1.00
sd(distractor_type1)                                                  1.00
sd(distractor_relatedness1)                                           1.00
sd(distractor_type1:distractor_relatedness1)                          1.00
cor(Intercept,distractor_type1)                                       1.00
cor(Intercept,distractor_relatedness1)                                1.00
cor(distractor_type1,distractor_relatedness1)                         1.00
cor(Intercept,distractor_type1:distractor_relatedness1)               1.00
cor(distractor_type1,distractor_type1:distractor_relatedness1)        1.00
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1) 1.00
                                                                      Bulk_ESS
sd(Intercept)                                                             1196
sd(distractor_type1)                                                      3274
sd(distractor_relatedness1)                                               2270
sd(distractor_type1:distractor_relatedness1)                              5148
cor(Intercept,distractor_type1)                                           7003
```

```
cor(Intercept,distractor_relatedness1)                                    6400
cor(distractor_type1,distractor_relatedness1)                             3813
cor(Intercept,distractor_type1:distractor_relatedness1)                   8256
cor(distractor_type1,distractor_type1:distractor_relatedness1)            4378
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     3664
                                                                       Tail_ESS
sd(Intercept)                                                             1877
sd(distractor_type1)                                                      2285
sd(distractor_relatedness1)                                               2232
sd(distractor_type1:distractor_relatedness1)                              2538
cor(Intercept,distractor_type1)                                           2739
cor(Intercept,distractor_relatedness1)                                    2283
cor(distractor_type1,distractor_relatedness1)                             2870
cor(Intercept,distractor_type1:distractor_relatedness1)                   2522
cor(distractor_type1,distractor_type1:distractor_relatedness1)            2769
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)     3027

~pic (Number of levels: 36)
                                                                       Estimate
sd(Intercept)                                                             50.48
sd(distractor_type1)                                                       3.29
sd(distractor_relatedness1)                                                4.05
sd(distractor_type1:distractor_relatedness1)                               4.66
cor(Intercept,distractor_type1)                                           -0.02
cor(Intercept,distractor_relatedness1)                                    -0.07
cor(distractor_type1,distractor_relatedness1)                              0.02
cor(Intercept,distractor_type1:distractor_relatedness1)                    0.01
cor(distractor_type1,distractor_type1:distractor_relatedness1)             0.01
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)      0.01
                                                                       Est.Error
sd(Intercept)                                                               7.04
sd(distractor_type1)                                                        3.33
sd(distractor_relatedness1)                                                 4.85
sd(distractor_type1:distractor_relatedness1)                                6.02
cor(Intercept,distractor_type1)                                             0.44
cor(Intercept,distractor_relatedness1)                                      0.42
cor(distractor_type1,distractor_relatedness1)                               0.45
cor(Intercept,distractor_type1:distractor_relatedness1)                     0.44
cor(distractor_type1,distractor_type1:distractor_relatedness1)              0.45
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)       0.45
                                                                       l-95% CI
sd(Intercept)                                                             38.55
sd(distractor_type1)                                                       0.10
```

```
sd(distractor_relatedness1)                                                  0.13
sd(distractor_type1:distractor_relatedness1)                                 0.09
cor(Intercept,distractor_type1)                                             -0.80
cor(Intercept,distractor_relatedness1)                                      -0.80
cor(distractor_type1,distractor_relatedness1)                               -0.80
cor(Intercept,distractor_type1:distractor_relatedness1)                     -0.79
cor(distractor_type1,distractor_type1:distractor_relatedness1)              -0.81
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)       -0.82
                                                                         u-95% CI
sd(Intercept)                                                               66.26
sd(distractor_type1)                                                        12.45
sd(distractor_relatedness1)                                                 17.31
sd(distractor_type1:distractor_relatedness1)                                22.23
cor(Intercept,distractor_type1)                                              0.79
cor(Intercept,distractor_relatedness1)                                       0.76
cor(distractor_type1,distractor_relatedness1)                                0.83
cor(Intercept,distractor_type1:distractor_relatedness1)                      0.81
cor(distractor_type1,distractor_type1:distractor_relatedness1)               0.82
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)        0.82
                                                                         Rhat
sd(Intercept)                                                            1.00
sd(distractor_type1)                                                     1.00
sd(distractor_relatedness1)                                              1.00
sd(distractor_type1:distractor_relatedness1)                             1.00
cor(Intercept,distractor_type1)                                          1.00
cor(Intercept,distractor_relatedness1)                                   1.00
cor(distractor_type1,distractor_relatedness1)                            1.00
cor(Intercept,distractor_type1:distractor_relatedness1)                  1.00
cor(distractor_type1,distractor_type1:distractor_relatedness1)           1.00
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1) 1.00
                                                                     Bulk_ESS
sd(Intercept)                                                            1416
sd(distractor_type1)                                                     3063
sd(distractor_relatedness1)                                              2460
sd(distractor_type1:distractor_relatedness1)                             3354
cor(Intercept,distractor_type1)                                          5605
cor(Intercept,distractor_relatedness1)                                   6052
cor(distractor_type1,distractor_relatedness1)                            4216
cor(Intercept,distractor_type1:distractor_relatedness1)                  7651
cor(distractor_type1,distractor_type1:distractor_relatedness1)           4811
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)    2911
                                                                     Tail_ESS
sd(Intercept)                                                            2411
```

```
sd(distractor_type1)                                                    1975
sd(distractor_relatedness1)                                             2119
sd(distractor_type1:distractor_relatedness1)                            2042
cor(Intercept,distractor_type1)                                         2823
cor(Intercept,distractor_relatedness1)                                  2960
cor(distractor_type1,distractor_relatedness1)                           3400
cor(Intercept,distractor_type1:distractor_relatedness1)                 2806
cor(distractor_type1,distractor_type1:distractor_relatedness1)          3033
cor(distractor_relatedness1,distractor_type1:distractor_relatedness1)   3190

Regression Coefficients:
                                        Estimate Est.Error l-95% CI u-95% CI
Intercept                                 754.98     23.31   709.81   799.08
distractor_type1                           10.47      9.13    -7.42    27.92
distractor_relatedness1                    20.30      5.10    10.67    30.34
distractor_type1:distractor_relatedness1   -7.63      9.81   -27.12    11.64
                                        Rhat Bulk_ESS Tail_ESS
Intercept                               1.01      718     1450
distractor_type1                        1.00     3230     3089
distractor_relatedness1                 1.00     5471     3110
distractor_type1:distractor_relatedness1 1.00     6702     2973

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   124.12      1.69   120.78   127.59 1.00     6086     2768

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
proper_model |>
  gather_draws(b_distractor_relatedness1, b_distractor_type1, `b_distractor_type1:distractor_
  # plot
  ggplot(aes(x = .value,
             y = .variable)) +
    stat_halfeye() +
    geom_vline(xintercept = 0,
               lty = "dashed") +
  labs(x = "posterior estimate for reaction times",
       y = "") +
  theme_minimal()
```




```r
p_direction(proper_model)
```

```
Probability of Direction

Parameter                                |     pd
-------------------------------------------------
(Intercept)                              |   100%
distractor_type1                         | 87.50%
distractor_relatedness1                  | 99.95%
distractor_type1:distractor_relatedness1 | 78.62%
```

Report comparing our reanalysis and the original analysis

In our final proper model, we kept the general build of the model of the original Mascelloni et al.(2021) paper. However, they did a frequentist model and we used a Bayesian one. We used weakly informative priors which we based on the general spread of the data (it covered nicely the range that we observed in the dataframe). For our slope prior, we had an agnostic prior of 0 difference between conditions.

We got very similar coefficient values to the original paper, which increased our confidence in the original results. The specific coefficients (with their respective credible intervals) are provided below.

Average reaction time across all condition combinations is 754 ms with the 95% credible intervals ranging 707-801 ms. When it comes to distractor types, the difference between categorical and mediated distractors, is likely (pd= 86.70%) to be positive, 10.45 ms, CI = [-7.97, 28.30], however both no difference and effect going the opposite direction are possible, too. For distractor relatedness, the difference between unrelated and related is 20.31 ms, CI [10.23, 30.29]. Pd of direction is 100% meaning that the effect is always expected to be positive and zero is impossible given our data, priors and model. The interaction between distractor type and relatedness shows a difference of -7.44ms with a CI of [-26.40, 11.45]. Pd of direction is 77.85% negative, looking at the plot, we see that zero, thus no interaction, is still likely as it is within the 50% interval.

In addition, the coefficients we got from the (relatively) flat model were also almost identical to both our final model and the Mascelloni et al., (2021) linear mixed effects model. However, we as Bayesians do not look at a p-value and make a yes/no significance decision, which is something the frequentist analysist does. But for the sake of comparison, we still computed p-values out of the probability of direction we got from the flat model, and obtained extremely similar values.

In the original paper, Mascelloni et al., (2021) claim that there is evidence for a mediated phonological-semantic effect. Reading their paper, there is no clear explanation for this claim or we were unable to understand it from their writing. Out of context, we managed to gauge that the effect that would support the cascade lexical retrival model would be for mediated related words to take longer reaction times in ms than mediated unrelated words (we do not touch SESOIs in this assignment yet). The reaction time for mediated related words is predicted by the model to be likely to be 767.62 ms whereas the mediated unrelated words 751.03 ms. This shows a result that favors the cascade model of lexical retrieval, which is what the original paper claimed. In addition, we see a promising interaction which the authors were expecting: The difference in reaction times between unrelated categorical distractors and related categorical distractors is 24.03 ms. The difference in reaction times between unrelated mediated distractors and related mediated distractors is also 16.59 ms.

We believe that our results show some evidence of the distractor relatedness making a difference in how fast (or slow) the participant was in naming a picture. All in all, we agree with the original authors.